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The last gasp of the Rogaland Igneous Complex, Norway: a palaeopole for the 920 Ma Tellnes intrusion

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SUMMARY

The Rogaland Igneous Complex (RIC) in southern Norway intruded into Sveconorwegian granulite crust beginning ∼930 Ma. Three massif anorthosite bodies, Egersund–Ogna, Helleren and Åna-Sira, were intruded some 10 Myr later by the Bjerkreim–Sokndal layered intrusion. The Garsaknatt leuconorite and the ilmenite-rich Tellnes norite, one of the youngest rock in the complex at ∼920 Ma, intrude the anorthosite or nearby country rock. Magnetic mineralogy and palaeomagnetic studies carried out on the Tellnes norite, the Garsaknatt leuconorite and the surrounding Åna-Sira anorthosite, indicate the magnetization of all three bodies are dominated by hemo-ilmenite carrying the remanence as a thermochemical remanent magnetization, although magnetite is present in some samples. The three bodies yield steep negative inclinations with northwesterly declinations (Tellnes, $I = -71.9°$, $D = 305.0°$, $\alpha_{95} = 10.6°$; Garsaknatt, $I = -73.1°$, $D = 312.7°$, $\alpha_{95} = 4.7°$; and Åna-Sira, $I = -81.2°$, $D = 326.3°$, $\alpha_{95} = 6.7°$). When combined with data from other bodies in the RIC, the older anorthosites have steeper inclinations, and higher palaeolatitudes, while the younger units have less steep inclinations and shallower palaeolatitudes by nearly 10°, indicating northward plate motion during cooling of the intrusions. Age of the remanence is difficult to determine precisely, however, best estimates are ∼910 Ma for the older anorthosites and ∼900 Ma for the younger intrusions. Although these differences are significant, a unified pole position (35.6° N, 215.1° E), combining all the 111 sites from the RIC, strongly supports the assumed position of southern Baltica in Rodinia at ∼900 MA.

Key words: Magnetic mineralogy and petrology; Palaeomagnetism; Rock and mineral magnetism.

1 INTRODUCTION

The location and spatial relationship of cratonic bodies in the past has long been of interest to geoscientists. From the 'fit' of Africa to South America to details of microplates, geological evidence has been pursued to investigate past continents and supercontinents. One of the most well known of supercontinents is Rodinia, where, at ∼1 Ga, all the ancient continental masses are suggested to be amalgamated into a supercontinent (Li *et al.* 2008; Merdith *et al.* 2017; Hoffman 1991; Martin *et al.* 2020). Yet, the exact positions of separate entities within the supercontinent are disputed (Karlstrom *et al.* 2001; Evans 2013; Evans *et al.* 2021). The only quantitative method to obtain position data of cratonic segments is to employ palaeomagnetism, where we can obtain not only palaeopoles for specific locations, but also the palaeolatitude at which cratons resided. Precisely dated, undeformed rocks with known geological histories, and information on the timing of magnetization are required to ensure accurate and realistic positions for continental blocks.

The position of Baltica, the ancient craton of present-day Scandinavia and north-eastern Russia, within the Rodinia supercontinent, is usually given a location bordering the northeastern most corner of present-day Laurentia (North America) at high southerly latitudes (Li *et al.* 2008), although other positions have been suggested (Evans *et al.* 2021). Most recently even the presence of Baltica within Rodinia has been questioned (Slagstad & Kulakov 2023). The pole path of Baltica during Rodinian time involves several hair-

*Deceased

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pin loops in earlier times (1.65–1.04 Ga), as well as complicated, not-well understood motion during the late Mesoproterozoic-early Neoproterozoic (1.05–0.85 Ga, Salminen *et al.* 2021; Kulakov *et al.* 2022).

In the southeastern most tip of Baltica is the Rogaland Igneous Complex (RIC), a sequence of anorthosites, a large layered intrusion, and associated dykes and smaller intrusive bodies. This complex was intruded into the deformed and metamorphosed rocks of Baltica at the end of the Sveconorwegian orogeny prior to the break-up of Rodinia (Bingen et al 2021). Previous palaeomagnetic work on the Egersund–Ogna anorthosite (Brown & McEnroe 2004) and the Bjerkreim–Sokndal (BKS) layered intrusion (Brown & McEnroe 2015) have determined the positions of these bodies in the mid-Tonian from dated undeformed rocks most with no, or only minor, evidence of later alteration. Such studies are able to shed light on details of supercontinent motions and alignments, and provide information on the position of Baltica within Rodinia at this time.

Here, we investigate one of the youngest known bodies in the Rogaland Igneous Province, the Tellnes ore body, an ilmenite-rich norite, currently mined for its ilmenite content, which intrudes the Åna-Sira anorthosite. The Tellnes is dated some 12 Myr younger than the initial anorthosite intrusions (Schärer *et al.* 1996). In addition, we report on magnetic signatures in the surrounding Åna-Sira anorthosite, as well as the separate but proximal Garsaknatt leuconorite. Palaeomagnetic data from these bodies plus previously published directions from the RIC will provide detailed information of the position of Baltica ∼900 Ma, and allow investigations into the changing position of the craton.

2 GEOLOGICAL AND GEOPHYSICAL SETTING

The Sveconorwegian Orogeny (Slagstad *et al.* 2017; Bingen *et al.* 2021) exposed in present-day southwestern Scandinavia, has traditionally been attributed to the complex collision of Baltica with Laurentia and Amazonia (Li *et al.* 2008; Bingen *et al.* 2021, and references therein) during the amalgamation of Rodinia. Bingen *et al.* (2021) describe the orogen as active in the late Mesoproterozoic to early Neoproterozoic (1065–920Ma) with several periods of high-pressure metamorphism, magmatism and deformation related to continental collision. Recent work in the region has suggested the orogeny may be better explained by wide continental backarc environment experiencing accretionary tectonics (Granseth *et al.* 2020; Slagstad *et al.* 2020). Under either explanation, the surrounding metamorphosed country rock, with youngest dates ∼980 Ma (Bingen & Van Breemen 1996; Bingen *et al.* 1998; Bingen & Stein 2003), represents the end of Sveconorwegian metamorphism in this part of southern Baltica. Orogenic activity, mainly extension and magmatism continues to −920 Ma (Slagstad *et al.* 2017; Bingen *et al.* 2021) with the emplacement of several large granitoids suites (Granseth *et al.* 2020).

The Neoproterozoic Rogaland Intrusive Complex (RIC) represents the last gasp of magmatism associated with the waning Sveconorwegian orogeny with the emplacement of the anorthosite–mangerite–charnockite igneous province. RIC has an estimated volume of some 5000 km^3 (Schärer *et al.* 1996), although modelling of offshore gravity and magnetic data indicate approximately 70 per cent more of the complex extends south into the Norwegian–Danish Basin (Olesen *et al.* 2004). Onshore the complex consists of three large 'massif' anorthosite bodies (Egersund–Ogna, Helleren and Åna-Sira), the BKS layered intrusion, and several smaller intrusive bodies including the Farsund charnockite and the Garsaknatt leuconorite (Fig. 1a). U–Pb zircon ages from the three large anorthosites are tightly clustered at 931 ± 2 Ma (Schärer *et al.* 1996) and samples from the quartz mangerite of the BKS yield an emplacement age 920 ± 6 Ma (Vander Auwera *et al.* 2011), both indicating intrusion some 50–60 Myr after the last deformation in the area. Additional ages on the Tellnes ore body and cross-cutting jotunite dykes are 920 Ma or younger (Schärer *et al.* 1996; Vander Auwera *et al.* 2011). This indicates the RIC was emplaced in two distinct pulses, one at ∼930 Ma including the massive anorthosites followed by a second pulse with the BKS layered intrusion and smaller units 10 Myr later at ∼920 Ma. Alternatively, Slagstad *et al.* (2022) suggest the anorthosites were emplaced over a long time period beginning around 950 Ma but remaining in the middle crust until later cooling around 915–900 Ma. Large xenoliths of anorthosite occur in the southwestern area of the BKS indicating that the anorthosites crystallized prior to the emplacement of the BKS intrusion (Paludan et al 1994).

The RIC has been of interest to geologists for over a century, for both the presence of natural resources (Korneliussen *et al.* 1999, 2001; Schiellerup *et al.* 2003) and for the still unsolved geological problems associated with anorthosites (Duchesne 1984; Ashwal & Bybee 2017). Reviews of existing literature (Duchesne 2001; Duchesne & Korneliussen 2003, and references within) and new geological maps (Marker *et al.* 2003; Marker 2013) provide excellent background for conducting detailed magnetic studies.

The Tellnes ore body intrudes into the Åna-Sira anorthosite in an elongated 'dyke-like' form (Fig. 1b), 2700 m long and 400 m wide, dipping 45° to the southwest (Krause *et al.* 1985). The body was first identified using aeromagnetic data in the late 1950s and is now one of the largest open-pit ilmenite mines in the world. The ilmenite norite, composed primarily of plagioclase (53 per cent), ilmenite (29 per cent), and orthopyroxene (10 per cent), with variable amounts of magnetite (0.7–2.5 per cent) has sharp contacts with the surrounding coarse-grained anorthosite (Krause *et al.* 1985). The body is associated with minor jotunite dykes, similar in composition to several larger dykes found in other parts of the RIC. The Tellnes body is crosscut by several splays of a dolerite dyke of the Egersund Dyke Swarm (Marker *et al.* 2003), dated at 616 ± 3 (U–Pb dating of baddeleyite by Bingen *et al.* 1998), 300 Myr younger than the norite. The Egersund Dyke Swarm is found across the RIC intruding the anorthosites in an ESE to WNW direction, unique from the RIC rocks by their mafic composition (Bingen *et al.* 1998).

Field relationships indicate the Tellnes ore body intruded the anorthosite, and a radiometric age of 920 ± 3 Ma based on U–Pb ID-TIMS dating of zircons (Schärer *et al.* 1996) for the Tellnes norite confirms the norite was emplaced some 10 Myr after the anorthosites, identifying it as one of the youngest bodies in the RIC. Origin of the ilmenite-rich deposit has been connected to emplacement of a crystal mush within fracture zones, or other weaknesses in the anorthosite (Krause *et al.* 1985; Diot *et al.* 2003) and to *in-situ* magma mixing (Robinson *et al.* 2003) or *in-situ* crystallization from an evolving magma (Charlier *et al.* 2006).

The RIC has been extensively studied using geophysical methods. A gravity survey of the area was made by Smithson & Ramberg (1979), and later a seismic reflection profile was collected from the Åna-Sira anorthosite to the western edge of the BKS intrusion (Deemer & Hurich 1997). Both studies concluded that the BKS

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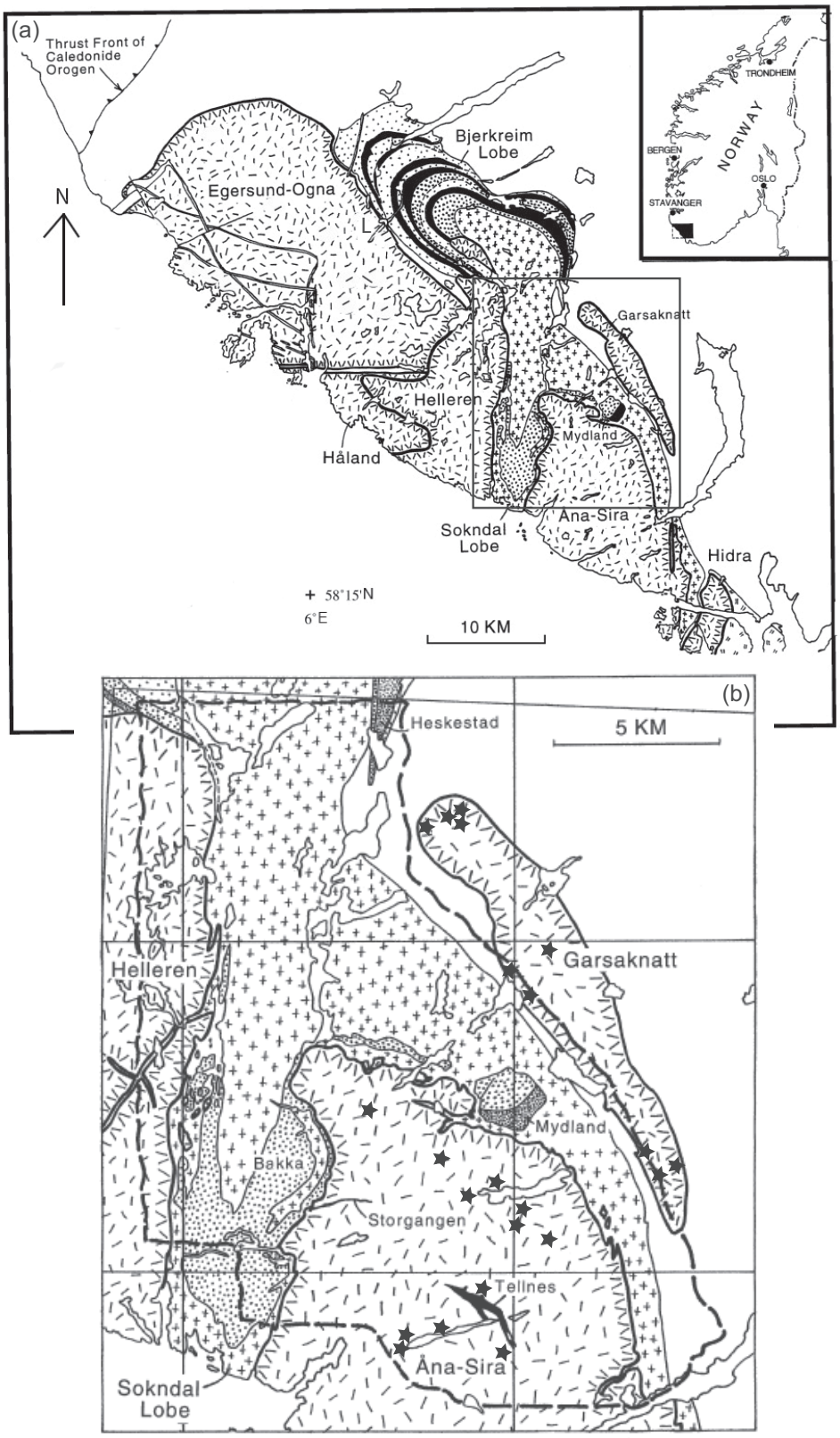


Figure 1. Geological and sample locations maps of the RIC modified from McEnroe et al 2001. Inset indicates location of RIC in southwestern most Norway. (a) Geological map of RIC showing the anorthosites (Egersund–Ogna, Håland-Helleran and Åna-Sira), the major layered intrusion (Bjerkreim–Sokndal with Bjerkreim and Sokndal lobes indicated), and the Garsaknatt leuconorite. L = location of Lomland Jotunite Dyke. Box indicates location of B map. (b) Enlargement of Åna-Sira anorthosite and the intruded Tellnes norite ore body. Palaeomagnetic sites in the Åna-Sira anorthosite and the Garsaknatt leuconorite are indicated by black stars; 10 sites (not shown) in Tellnes norite are located in the active Tellnes mine area (dark narrow outcrop in the lower centre of the map).

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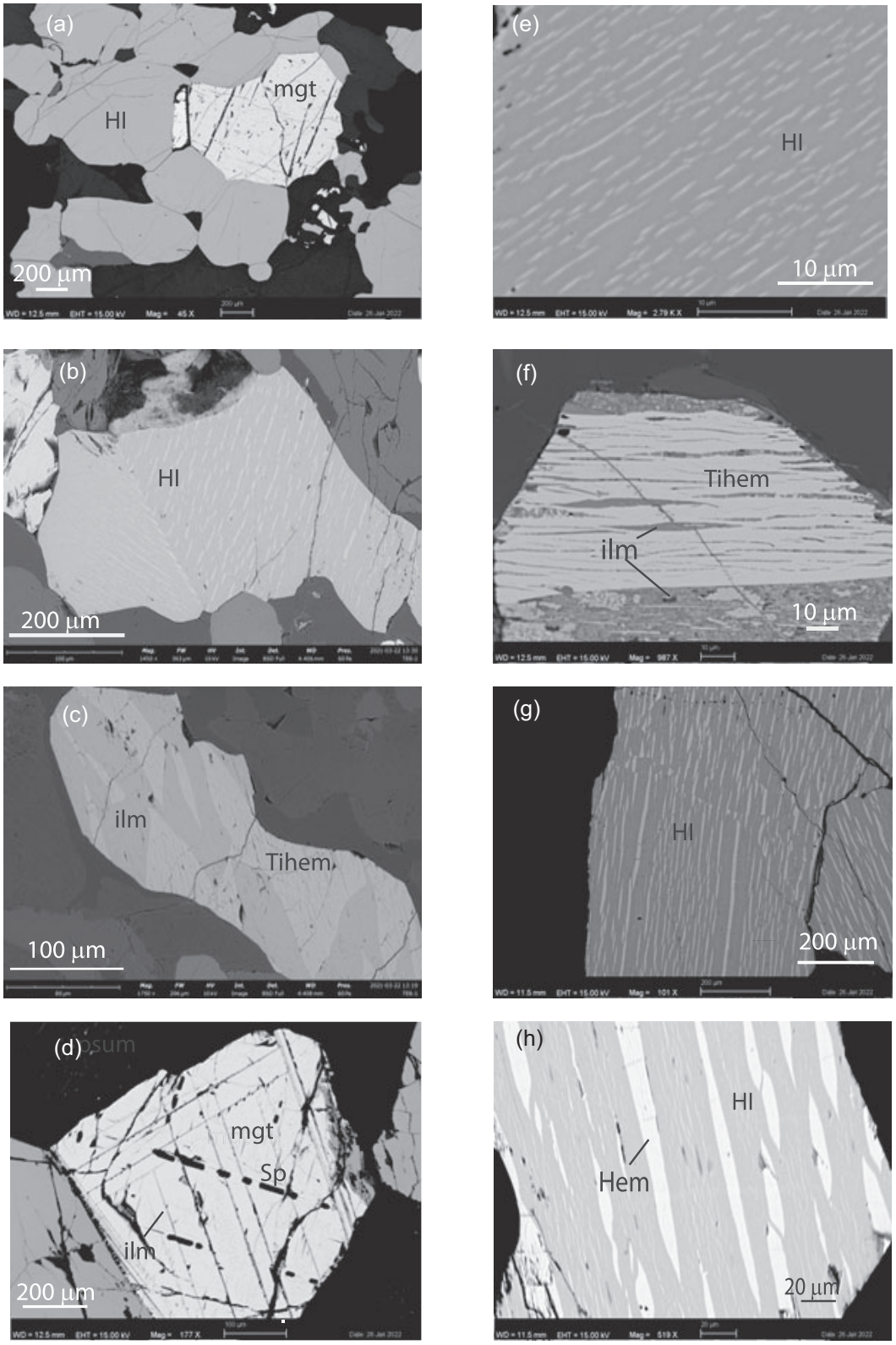


Figure 2. (a)–(d) Tellnes norite. (a) Hemo-ilmenite (HI) with haematite lamellae and large magnetite grain with spinel (sp) and ilmenite (ilm) oxy-exsolution lamellae. (b) Hemo-ilmenite with fine haematite lamellae. (c) Titanohaematite grain with large ilmenite lamellae. (d) Large magnetite grain with ilmenite and spinel exsolution. (e) and (f) Åna Sira anorthosite. (e) Very fine haematite lamellae in hemo-ilmenite. (f) Titanohaematite with ilmenite exsolution lamellae. Ilmenite has altered partly to rutile. (g) and (h) Garsaknatt leuconorite. (g) Large hemo-ilmenite grain with abundant haematite lamellae. (h) Hemo-ilmenite with two generations of haematite lamellae.

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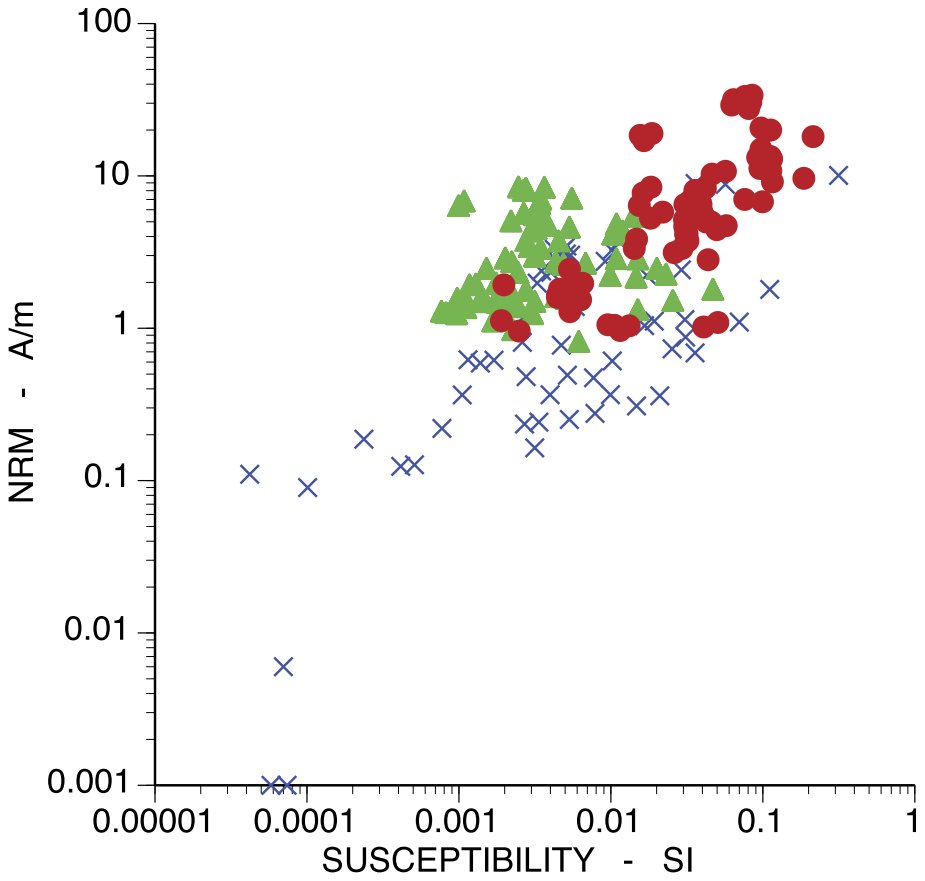


Figure 3. Logarithmic plot of remanent magnetization versus susceptibility for samples from the Tellnes norite (dots), the Åna-Sira anorthosite (triangles) and the Garsaknatt leuconorite (Xs).

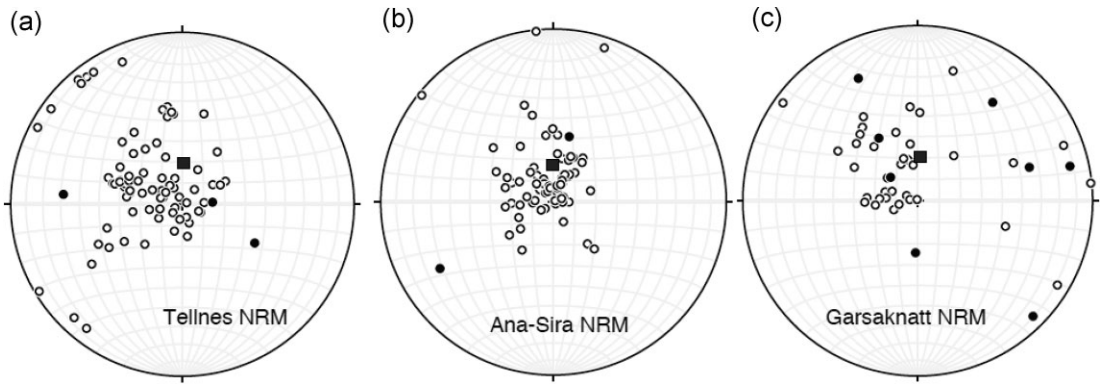


Figure 4. Stereographic projections of NRM of all individual samples (in geographic coordinates). (a) Tellnes norite, (b) Åna-Sira anorthosite and (c) Garsaknatt leuconorite. Open circles indicate projections on the upper hemisphere (negative inclinations); black circles indicate projections on the lower hemisphere (positive inclinations). Large filled square is the present earth's field for southern Norway.

intrusion has a depth of 4–5 km. Because the BKS intrusion intrudes into the anorthosites this provides a minimum depth for the enclosing anorthosites.

Numerous magnetic airborne fixed-wing (Olesen *et al.* 2015), helicopter (Rønning 1995), and drone (Lee *et al.* 2023) surveys have been flown in the RIC region to aid in the geological interpretation, and to assess for mineral potential, for ilmenite, magnetite and apatite reserves. The Sokndal lobe of the BKS and adjacent Åna-Sira anorthosite, which the Tellnes body intrudes, were the focus of a high-resolution helicopter magnetic survey. This survey was acquired at 40 m height and line spacing of 0.1 km and provides excellent constraints on the geometry of the Tellnes norite (Rønning 1995). Regional geophysical overviews are given by Nasuti *et al.* (2015) and Olesen *et al.* (2015). McEnroe (1997) and McEnroe *et al.* (1996, 2001) discussed the magnetic and chemical properties of the Tellnes norite and Åna-Sira anorthosite. Detail AMS studies by Biedermann *et al.* (2016) and Bolle *et al.* (2000) provides information on the fabric and magma emplacement in the BKS and

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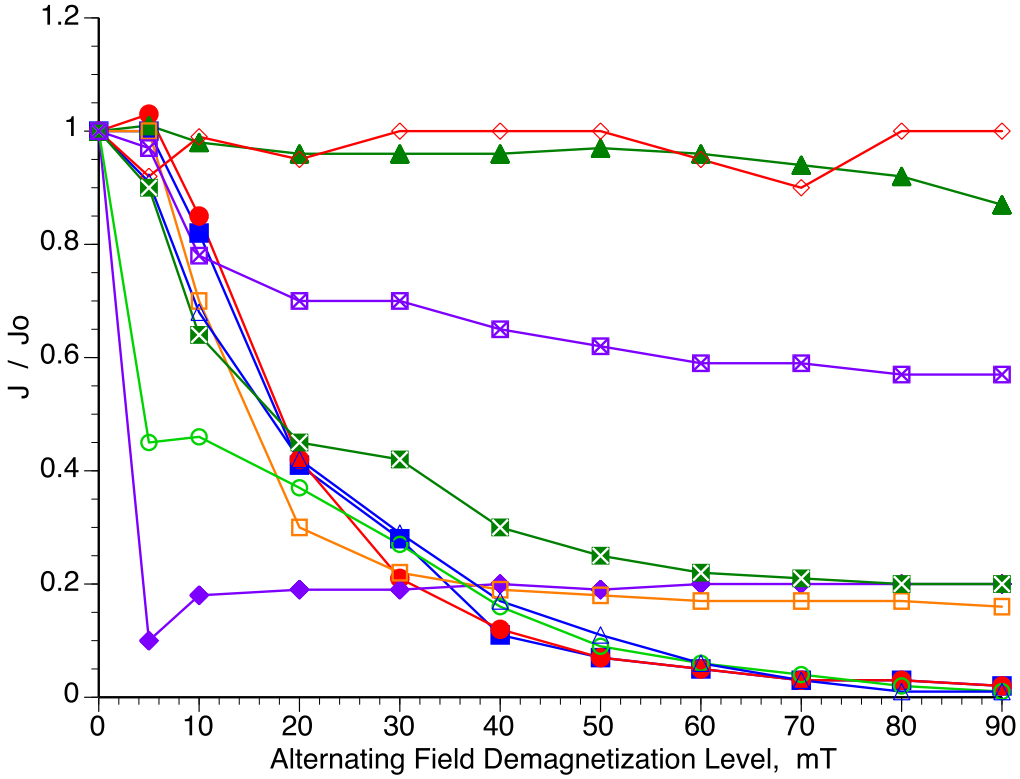


Figure 5. Plot of normalized remanence (*J*/*Jo*) to alternating field demagnetization steps from 0 to 90 mT of selected samples from Tellnes norite samples.

surrounding anorthosites. Later, Biedermann &. McEnroe (2017) evaluated the effects of magnetic anisotropy on total magnetic field anomalies in the BKS.

3 MAGNETIC MINERALOGY

3.1 Tellnes norite ore body

The magnetic mineralogy and petrology of the Tellnes norite is described in detail by McEnroe *et al.* (1996, 2001). Robinson *et al.* (2001) reported additional electron microprobe analyses of hemo-ilmenite and magnetite from samples from our palaeomagnetic sites, which expand further the features observed in reflected-light microscopy. Here, we summarize the relevant magnetic mineralogy and oxide microstructures found in the Tellnes norite from these studies and add in new observations from scanning electron microscopy (SEM). These images are shown in Figs 2(a)–(d).

The Tellnes norite is composed of coarse- grained plagioclase, orthopyroxene, clinopyroxene, biotite, olivine, hemo-ilmenite, magnetite and pyrite. In parts of the mine the hemo-ilmenite content is up to 29 per cent. In the more primitive samples hemo-ilmenite grains are a ferrian ilmenite, ($Im_{96\text{-}99}$) which contains abundant fine haematite exsolution lamellae parallel to (0001) with submicrometre second-generation haematite lamellae, also parallel to (0001). Pleonaste spinel (along the $MgAl_2O_4$–$FeAl_2O_4$ join with subordinate Fe^{3+}) commonly exsolved as needles from more primitive hemo-ilmenite grains. In these samples (sites TE 3 and TE 11) ilmenite grains contain up to 20 per cent geikielite component ($MgTiO_3$). The more evolved hemo-ilmenite grains analysed from site BK 79 generally contain less haematite lamellae, and lower geikielite component, however, richer in Ti component. Commonly where magnetite and ilmenite are adjacent grains, there is a magnetite-spinel-ilmenite symplectite at the border between the two grains showing a subsolidus re-equilibration. The magnetite grains have minor (111) oxidation-exsolution of ilmenite lamellae, and small (100) spinel lamellae. Internal to some orthopyroxene grains are ilmenite exsolution lamellae with submicron exsolution of haematite lamellae. In the subordinate clinopyroxene grains there are very-fine plates or rods of ilmenite and magnetite with ilmenite oxidation exsolution, parallel to (010). The blades are typically 10–50 μm long and only a micrometre wide.

The abundant haematite lamellae (< 1 μm) in the hemo-ilmenite were not large enough to obtain good analytical results. However, the lowest ratio 2Ti/(2Ti + R3+) reported by Robinson *et al.* (2001) at about 0.3 which is near the stable R3c haematite composition of 0.28 on the Burton ilmenite-haematite phase diagram (see fig. 3, in McEnroe *et al.* 2001). The maximum Ti analyses indicate the compositions of ilmenite hosts free of haematite lamellae. Ilmenite is considered a primary magmatic precipitate whereas magnetite likely crystallized later from an interprecipitate liquid. Large magnetite grains are observed in thin sections examined from the norite. These grains are usually 100 s of micrometres wide and many more than 1 mm in size (Fig. 2a). In the more primitive samples, the common microstructure in the magnetite grains is formed by spinel exsolution, as small and large blades and needles. Minor oxidation-exsolution of ilmenite lamellae occurs in magnetite, which often contains spinel needles at the edges of the ilmenite lamellae (Figs 2a and d).

New SEM observations on a thin section from site TE 8, associated with the late stage evolution of the magmatic intrusion, reveals features not reported in earlier studies. Along with magnetite, both hemo-ilmenite and titanohaematite, with il-

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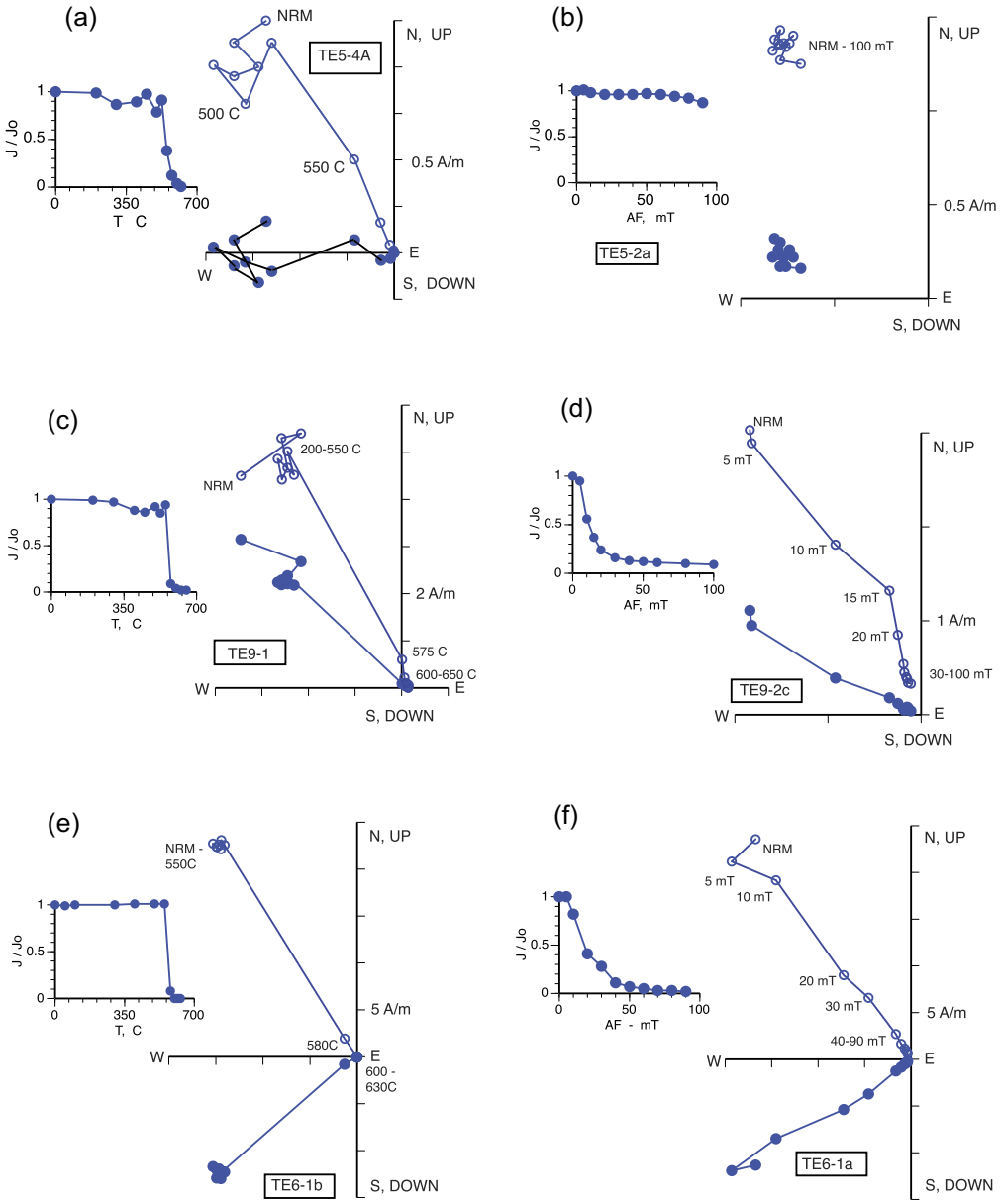


Figure 6. Vector endpoint diagrams for thermal (left-hand column) and alternating field (right-hand column) demagnetization of samples from the Tellnes norite (in geographic coordinates). Open circles are projections on the vertical plane; solid circles are projections on the horizontal plane. Insets are plots of normalized magnetization to demagnetization level.

menite exsolution lamellae, are abundant in this thin section (Fig. 2c). In Rogaland titanohaematite has only previously been reported in the Egersund–Ogna anorthosite (Brown & McEnroe 2004) and is also known to occur in some of the surrounding metamorphic rocks. This may indicate some crustal contamination to the norite dyke from the surrounding country rock.

3.2 Åna-Sira anorthosite

The Åna-Sira anorthosite is composed of millimeter size plagioclase grains, minor orthopyroxene and accessory hemo-ilmenite, and rare magnetite grains (Figs 2e and f). Hemo-ilmenite is the dominant oxide and a primary phase that crystallized as discrete grains of ferrian ilmenite which, with cooling to below solvus temperatures (520 °C), exsolved haematite lamellae (Fig. 2f). Inclusions of hemo-ilmenite are also found in the plagioclase grains. Thin section observations on petrographic and SEM show the hemo-ilmenite grains contain two generations of haematite exsolution lamellae, ranging from a few micrometre to submicrometre size. Hemo-ilmenite-rich cumulates in the Åna-Sira anorthosite have been studied by transmission electron microscope and the haematite lamellae were found to continue down to a few nanometers in thickness (McEnroe *et al.* 2002). These very fine scale exsolution features in the hemo-ilmenite with nanoscale lamellae of haematite indicate the remanent magnetization was acquired by cooling from high temperatures, after emplacement of the anorthosites. Locally in some hemo-ilmenite grains, there is evidence for a late oxidation by the breakdown of ilmenite to haematite + rutile. Overall, the Åna-Sira samples show oxide mineralogy, with no to at most minor alteration.

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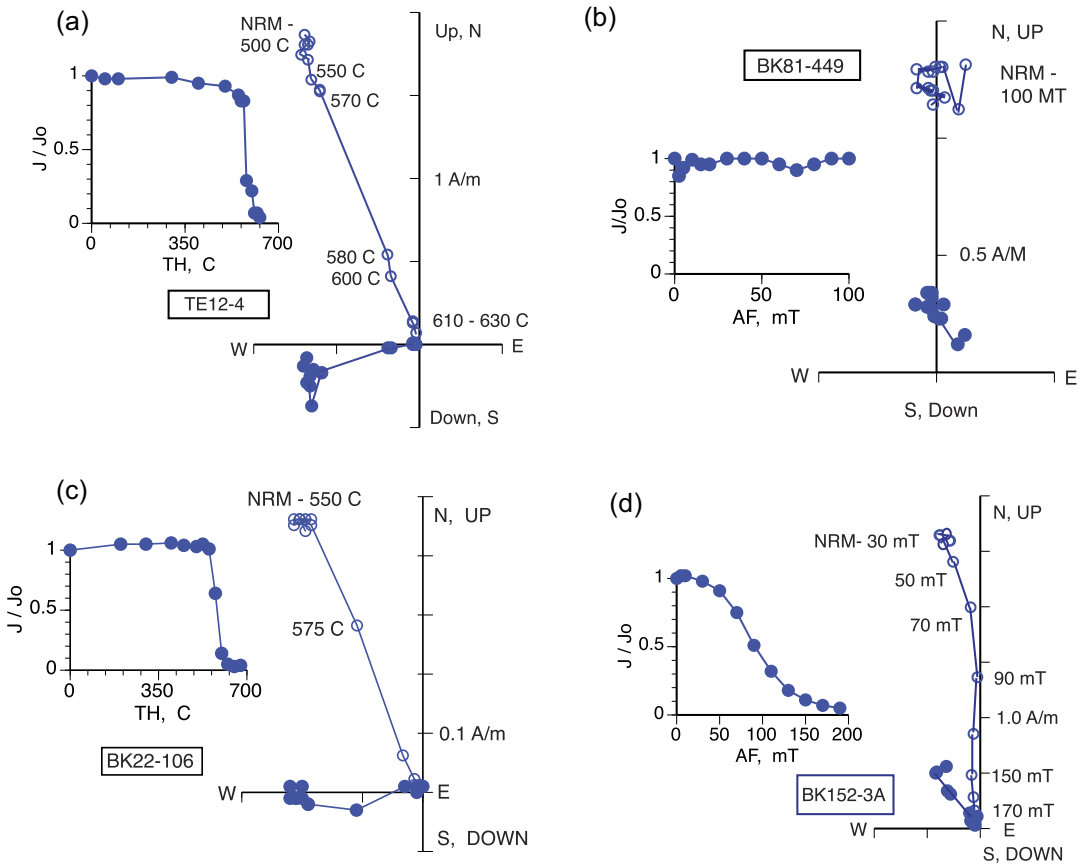


Figure 7. Vector endpoint diagrams (in geographic coordinates) for thermal (left-hand column) and alternating field (right-hand column) demagnetization for samples from (a) and (b) the Åna-Sira anorthosite and (c) and (d) the Garsaknatt leuconorite. Symbols as in Fig. 6.

3.3 Garsaknatt leuconorite

The Garsaknatt leuconorite is the least studied intrusion in Rogaland. It has been reported as a leuconorite, and more recently as an anorthosite (Marker *et al.* 2003). The difference is the amount of plagioclase. Because we have not determined the silicate ratios of plagioclase and orthopyroxene we will continue to use the older classification of leuconorite.

The Garsaknatt leuconorite is mainly composed of plagioclase, and lesser amount of orthopyroxene. The magnetic oxides are hemo-ilmenite (Fig. 2g) and magnetite. Hemo-ilmenite is present in all our thin sections, whereas discrete magnetite grains are absent in some sections. The silicate and oxide grains are generally millimetre in size. The hemo-ilmenite is similar to that reported from the Tellnes norite with abundant fine haematite exsolution parallel to (0001). However, only one sample was observed to have subsequent generation of haematite lamellae in the ilmenite (Fig. 2h). It is possible more abundant lamellae at the nanometer scale may be present. The amount and size of magnetite grains in the Garsaknatt are variable, with some samples having only rare magnetite grains and others with more than1 per cent of magnetite. Magnetite has minor oxidation-exsolution lamellae of ilmenite and only minor spinel exsolution. The abundance of ilmenite lamellae and spinel exsolutions in is less than found in magnetite grains from the Tellnes norite.

4 MAGNETIC MEASUREMENTS

4.1 Sampling

Samples from the Tellnes norite and surrounding rocks were initially collected in 1995 as part of a large project undertaken by the Norwegian Geological Survey to improve interpretation of aeromagnetic signature related to ilmenite ore bodies. Much of the early sampling involved collecting oriented blocks within the open-pit working Tellnes mine proper, as well as a cross-cutting younger Egersund dyke and the surrounding Åna-Sira anorthosite rocks. Blocks were oriented in place with a compass aided by backsitting to topographic features. Four to six oriented cores were drilled from each block, and cores cut into 2.5 cm specimens. Additional samples were collected in 2001, 2003 and 2018, primarily as samples drilled in the field with a gasoline-powered rock drill, and oriented with magnetic and sun compass. The Garsaknatt leuconorite, which is exposed to the east of the main RIC, and a large dyke of jotunite composition, the Lomland dyke which intrudes the Egersund–Ogna anorthosite and the BKS layered intrusion (Fig. 1a), were also sampled, all by drilling on location. Sites are shown on Fig. 1(b). Due to inadequate rock exposures it was not possible to collect detailed contact samples to provide for a complete baked contact test for either the Lomland or Egersund dykes.

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Table 1. Palaeomagnetic data from sites in Tellnes norite, Åna Sira Anorthosite and Garsaknatt Leuconorite.

Site	SUSC	NRM	N/n	INC	DEC	K	a-95	PLAT	PLON
Tellnes	Norite								
BK 79	0.1050	10.398	7/7	−66.6	312.7	83	6.7	−24.3	218.3
BK 80	0.1012	12.743	6/6	−59.7	290.1	108	6.5	−24.6	238.1
BK 81	0.0110	1.234	8/8	−74.6	29.4	102	5.5	−31.7	170.3
TE 3	0.0409	7.2	8/8	−63.2	267.6	110	5.3	−37.9	250.6
TE 4	0.0291	4.22	7/7	−65.6	344.3	101	6.0	−16.8	197.4
TE 5	0.0050	1.71	9/9	−50.2	274.9	186	3.8	−23.6	255.2
TE 6	0.0737	30.18	7/10	−82.3	147.2	153	4.9	−69.5	162.6
TE 8	0.0715	7.3	10/11	−73.1	318.6	154	3.9	−31.5	210.2
TE 9	0.0431	2.71	8/8	−63.4	299.2	194	4.0	−24.9	229.3
TE 11	0.0154	9.77	5/5	−72.8	340.6	170	5.9	−27.6	197.8
Mean			10/10	−71.9	305.0	22	10.6	−33.3	218.9
			10/10			8	17.7	−34.1	216.4
Åna Sira	Anorthosite								
TE 1	0.0139	3.000	6/6	−72.8	240.8	51	8.7	−59.1	250.0
TE 2	0.0025	5.67	3/3	−67.2	286.2	375	6.4	−33.9	234.5
TE 10	0.0017	1.56	4/4	−78.1	288.7	190	6.7	−46.0	218.4
TE 12	0.0013	1.46	6/7	−75.1	276.1	58	8.2	−46.5	229.1
TE 43	0.0024	1.30	3/3	−81.2	339.7	221	8.3	−41.9	194.3
TE 44	0.0060	2.44	3/4	−83.2	33.7	143	10.3	−46.6	175.7
TE 46	0.0036	4.23	6/6	−63.6	8.8	67	8.3	−13.8	180.1
TE 47	0.0015	2.55	4/4	−78.6	345.0	993	2.5	−36.9	193.4
TE 48	0.0038	3.67	5/5	−76.2	327.1	65	9.6	−34.8	203.4
TE 49	0.0055	5.65	5/5	−78.0	41.1	191	5.6	−39.0	167.1
TE 50	0.0142	3.64	5/5	−82.4	55.9	196	5.5	−48.3	167.7
TE 54	0.0056	5.81	4/4	−76.3	334.3	245	5.9	−33.9	199.6
MEAN			12/12	−81.2	326.3	44	6.7	−41.9	195.4
			12/12			14	12.2	−42.8	199.3
Garsaknatt	Leuconorite								
BK 21	0.0041	0.30	5/6	−67.5	319.4	39	14.9	−23.8	213.0
BK 22	0.0031	O.32	4/4	−67.5	282.0	155	7.4	−36.1	236.4
BK 23	0.0057	1.70	5/5	−75.5	299.8	27	14.9	−39.7	217.1
BK 24	0.0002	0.14	2/2	−78.8	321.5	xx	xx	−39.8	215.4
BK51	0.0011	0.12	5/5	−65.5	302.8	24	14.3	−23.0	218.7
BK 52	0.0152	2.71	6/6	−72.1	292.0	63	8.5	−37.6	225.4
BK 53	0.0265	3.07	7/7	−71.1	332.7	26	12	−26.2	202.8
BK 54	0.0595	2.20	6/6	−69.1	343.3	139	5.7	−21.9	196.8
BK 55	0.0515	0.97	5/5	−72.4	323.9	56	10.3	−29.6	207.3
BK 152	0.0045	2.89	5/5	−81.4	315.6	572	2.6	−45.0	203.1
Mean			10/10	−73.1	312.7	105	4.7	−32.9	213.5
			10/10			39	7.8	−32.8	213.3
Jotunite dyke									
BK 145	0.0548	0.37	5/6	−69.3	327.9	24	14.4	−24.5	206.7
BK 146	0.0791	1.01	3/6	−74.4	270.9	357	65.3	−47.8	232.6
BK 147	0.0600	0.31	4/6	−77.9	298.7	56	10.7	−43.3	214.4
Mean			3/3	−75.2	302.3	52	14.0	−38.6	216.4
			3/3			26	24.8	−39.1	216.6
Egersund dyke									
TE-DK1	0.0008	0.01	6/6	70.3	113.0	60	8.7	34.9	47.2
TE-7	0.0206	1.34	7/7	58.7	27.5	196	4.3	64.2	131.4

Notes. SUSC, magnetic susceptibility, in SI; NRM, natural remanent magnetization in Am^{-1}; *N*/*n*, Number of samples used in calculation/number of samples measured; INC, inclination, in degrees; DEC, declination, in degrees; K, precision parameter; a95, radius of 95 per cent cone of confidence; PLAT, Latitude, in degrees, of virtual geomagnetic pole; and PLON, longitude, in degrees, of virtual geomagnetic pole and mean pole.

4.2 Methods

Susceptibility measurements were made in the Petrophysics laboratory at NGU and using a Sapphire Instruments Susceptibility Bridge or a Bartington Susceptibility Meter at UMass. Initial natural remanent magnetization (NRM) was measured on all samples, using the JR5-9 Hz spinner magnetometer at NGU or the 2 G SQUID magnetometer at the University of Massachusetts. Initial demagnetization measurements were made on selected samples using demagnetization equipment at NGU. Due to the high NRM intensities detailed demagnetization on a full suite of samples was made using a Molspin Alternating Field demagnetizer and an ASC thermal

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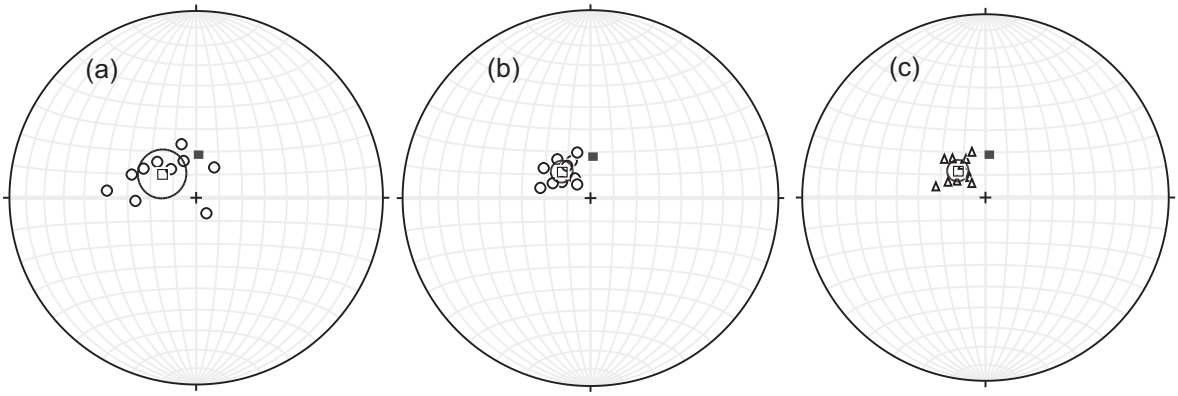


Figure 8. Equal-area plots of site mean directions (in geographic coordinates) for the (a) Tellnes norite, (b) Åna-Sira anorthosite and (c) Garsaknatt leuconorite. Open circles are site mean directions, open square is mean direction for the body; circle is 95 per cent confidence around the mean. Filled square is the present earth's field for southern Norway.

Table 2. Palaeomagnetic directions and poles for RIC units.

Unit	Rock type	Sites/units	Inc	Dec	k	α_{95}/A_{95}	Pole lat	Pole long	Palaeolat	References
Tellnes	Norite	10	−71.9	305.0	22	10.6	−33.3	218.9	−56.8	This paper
					8	17.7	−34.1	216.4		
Garsaknatt	Leuconorite	10	−73.1	312.7	105	4.7	−32.9	213.5	−58.7	This paper
					39	7.8	−32.8	213.3		
Bjerkreim–Sokndal	Layered Intrusion	66	−73.5	303.4	24	3.7	−35.9	217.9	−59.4	Brown & McEnroe (2015)
					10	5.9	−38.4	217.0		
MEAN—Younger Units		86	−73.3	304.7	26	3.0	−35.7	218.1	−59.4	This paper
					11	4.9	−37.2	216.5		
Egersund–Ogna	Anorthosite	13	−80.1	325.9	73	4.9	−41.5	200.3	−70.8	Brown & McEnroe (2004)
					22	9.1	−42.1	200.4		
Åna-Sira	Anorthosite	12	−81.2	326.3	44	6.7	−41.9	195.4	−72.8	This paper
					14	12.2	−42.8	199.3		
MEAN—Older units		25	−80.5	326.2	57	3.9	−42.2	199.9	−71.5	This paper
					18	7.1	−42.5	199.8		
All RIC-units		5	−76.1	312.1	264	4.7	−37.5	210.8	−63.7	This paper
					73	9.0	−37.6	209.7		
All RIC—sites		111	−75.1	307.8	28	2.6	−35.6	215.1	−62.0	This paper
					11	4.2	−38.7	212.8		This paper

Notes. Inc and Dec are palaeomagnetic inclination and declination, in degrees; k is the precision parameter; α_{95}/A_{95} are 95 per cent confidence circles about mean directions/mean poles; Pole Lat and Pole Long are positions of mean palaeopoles, in degrees; and Palaeolat is the palaeolatitude calculated from the mean direction.

demagnetizer at the University of Massachusetts. In all cases, alternating field demagnetization was made in 10–12 steps to 100 mT, and thermal demagnetization in 8–12 steps up to 675 °C. Magnetic directions were calculated using principle component analysis procedures (Kirschvink 1980) with at least three points determining linear fits, with maximum angular deviations less than 10° and more often less than 5°.

4.3 Rock magnetic properties

4.3.1 Magnetic susceptibility

To further delineate properties of the Tellnes norite ore body and compare them to the surrounding anorthosite, additional susceptibility measurements were made on all ore samples and the near-by Åna-Sira anorthosite in the collection. 72 samples from the Tellnes norite showed a wide range in susceptibility from 0.002 to 0.209 SI, with a mean value of 0.049 SI (Fig. 3). Many of the samples have susceptibilities greater than 0.01 SI reflecting the presence of magnetite in much of the ore body. However, it is notable that a few samples have much lower susceptibility derived from parts of the Tellnes where hemo-ilmenite is the only iron oxide present. At least one-third of the mine samples have susceptibility values greater than 0.05 SI indicating at least 1.7 per cent magnetite present.

The palaeomagnetic anorthosite samples from the Åna-Sira anorthosite ($N = 38$) have a mean value for susceptibility an order of magnitude lower (0.005 SI) than the Tellnes norite, with a range from 0.00078 to 0.04726 (Fig. 3a). The majority of samples have values below 0.005 SI that are similar to those reported in Brown & McEnroe (2008) on 110 Egersund–Ogna, Åna-Sira and Helleren anorthosite samples. These lower susceptibility values agree with the observed iron oxides, with only minor or no magnetite grains in the anorthosites, and a preponderance of hemo-ilmenite oxides. Garsaknatt leuconorite samples show a very wide

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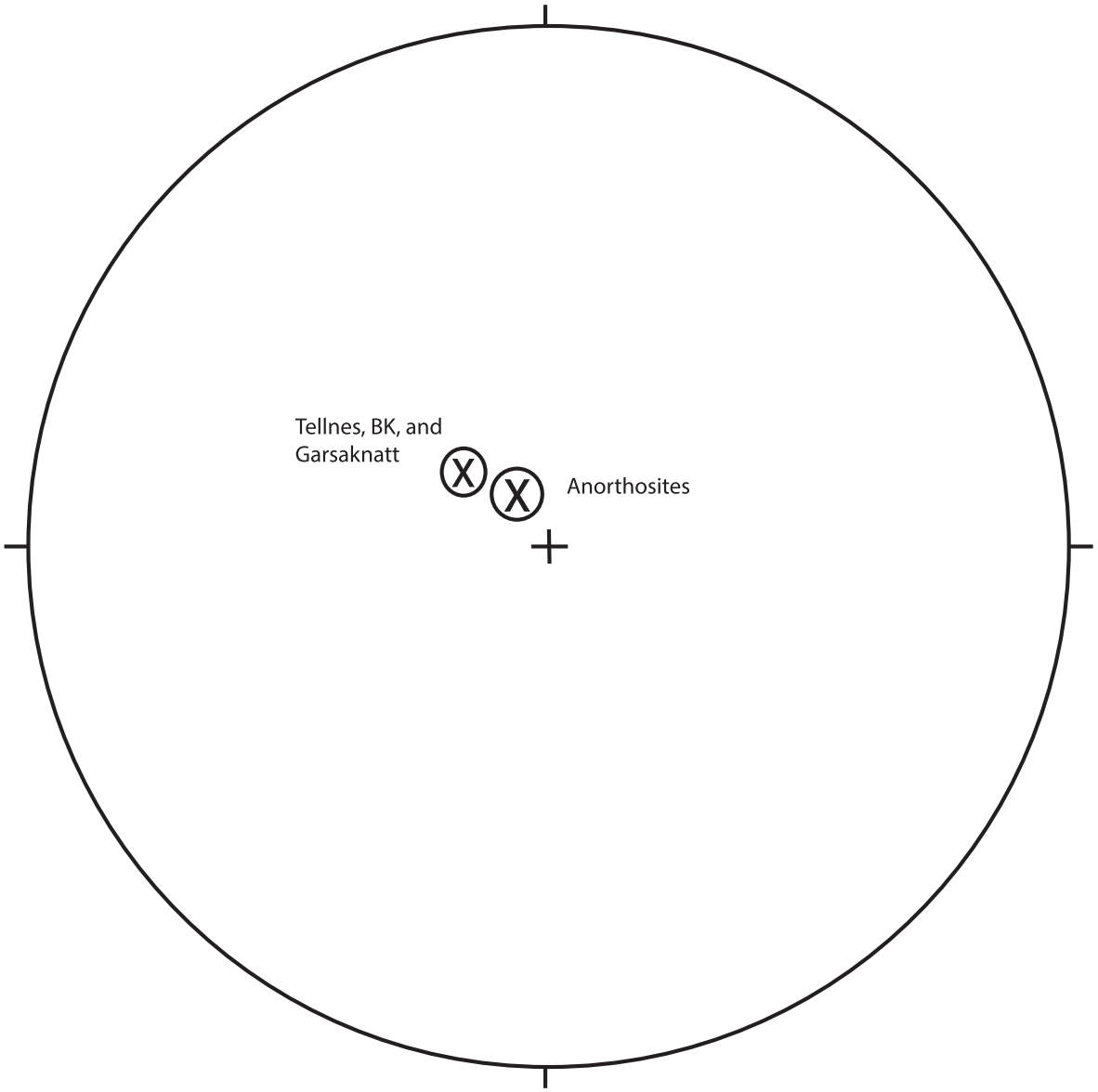


Figure 9. Equal-area plot of two group means from sites in the RIC. Grouping determined by age, with the older anorthosites (Egersund–Ogna and Åna Sira) separated from the younger units (BKS intrusion and Tellnes norite), with the undated Garsaknatt leuconorite included due to direction similarities. Open circles are mean directions projected on the upper hemisphere.

range of susceptibility values, from greater than 0.1 SI to less than 0.0001 SI, as observed on Fig 3. It appears that this leuconorite, not directly related to the Åna-Sira anorthosite spatially, is much more heterogeneous in its oxide composition than the other three RIC anorthosites. This difference is also observed in polished sections with both the size and composition of oxide grains being highly variable.

4.3.2 Natural remanent magnetization and Koenigsberger ratio (Q)

NRM has been measured on all the samples collected from the Tellnes norite as well as the surrounding Åna-Sira anorthosite. Remanence of the Tellnes norite samples is high, with a mean value of 9.30 A m^{-1} and a range of 0.95 to 33.09 A m^{-1} (Fig. 3). Directions from the NRM measurements are predominantly steep negative inclinations with west and northwest declinations (Fig. 4). Åna-Sira samples have magnetization with an average value of 3.53 A m^{-1} and a range of 0.83 to 8.50 A m^{-1}, which is quite remarkable considering the remanent magnetization is held in hemo-ilmenite, which commonly makes up less than 2 per cent of the rock. The hemo-ilmenite usually contains less than 8 per cent lamellae; therefor the total haematite content of the anorthosite samples is less than 0.02 per cent. The anorthosite samples have even steeper negative inclinations, and similar west to northwest declinations (Fig. 4). As with susceptibility, the NRM values from the Garsaknatt leuconorite have a wide distribution, with many of the samples having remanence less than the other two bodies, but also some higher values up to 10 A m^{-1} (Fig. 3). Directions from the Garsaknatt leuconorite are similar to the other bodies with predominantly negative inclinations, however, ∼30 per cent of samples have positive and/or shallow inclinations and scattered declinations.

The Koenigsberger ratio (Q) is derived as a ratio of remanent magnetization to induced magnetization; Q ratios for these rocks

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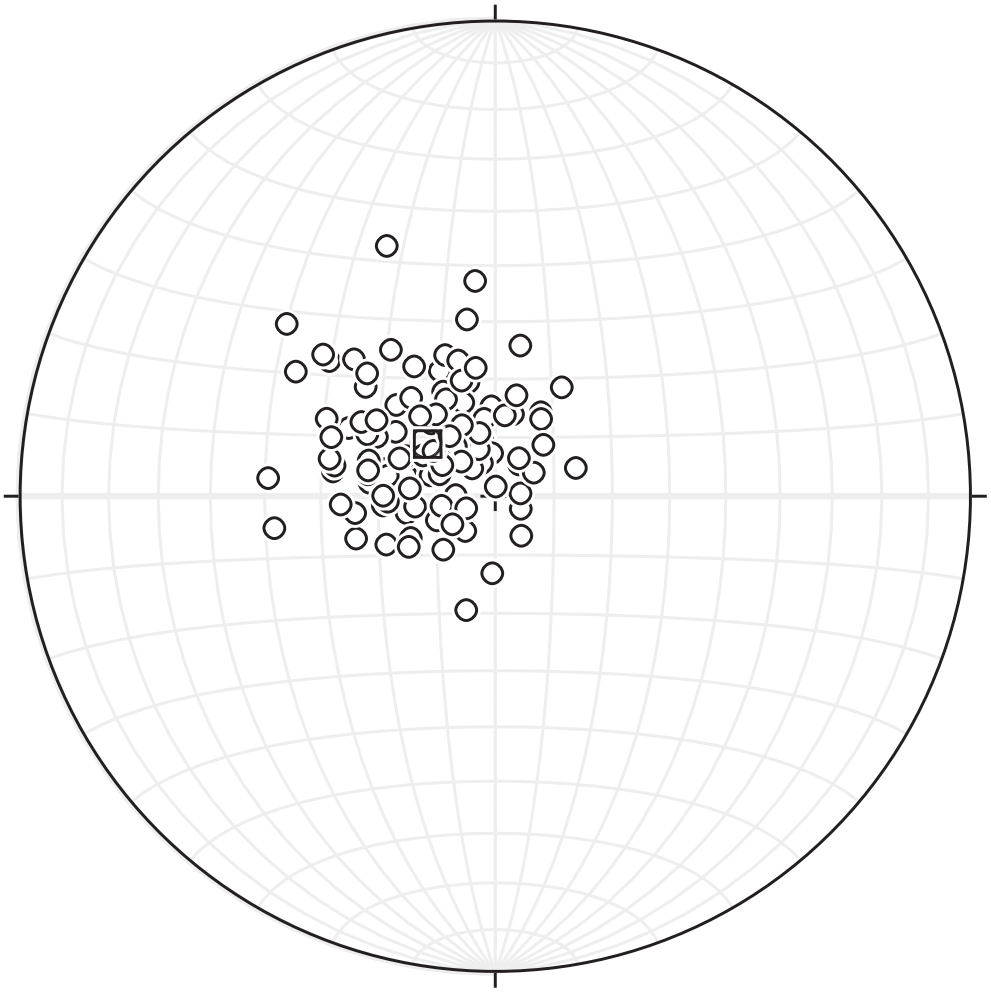


Figure 10. Equal-area plot of site mean directions (in geographic coordinates) for all samples from the RIC, as described in the text. Open circles are projections onto the upper hemisphere. Small box is location of mean direction for all 111 sites.

Table 3. Palaeomagnetic poles from southern Scandinavia.

Study	Age∗ (Ma)	Sites	Pole lat	Pole long	*A*95	References
Central Telemark	∼1000	23	−17.4	229.87	6.5	Kulakov *et al.* (2022)
Dykes, South Sweden	951–935	12	−2.6	239.6	5.8	Gong *et al.* (2018)
Dykes, South Sweden	935–946	11	−0.5	241	6	Elming *et al.* (2014)
Bamble Terrane	950–900	(191)	−32	231		Piper (2009)
Telemark Terrane	950–900	(151)	−34	239		Piper (2009)
West Telemark	900	9	−46.2	209.4	12.0	Kulakov *et al.* (2022)
RIC—Older	910	25	−42.5	199.8	7.1	This paper
RIC—Younger	900	86	−37.2	216.5	4.9	This paper
RIC	900	111	−38.6	212.8	4.2	This paper
'Anorthosite Complex'		76	−44.1	213.0	3.5	Stern & Piper (1984) (Recalculated)

Notes. AGE∗, age of remanence as reported in original paper; Sites, number of sites reported used in pole calculation; parentheses indicate number of components as defined in Piper (2009); other headings are in Table 1.

are all greater than 1.0. The Tellnes samples have an average Q ratio of 6.2, with the Garsaknatt leuconorite having an average of 7.8. The Åna-Sira anorthosite, with its low susceptibilities but relatively high remanence, averages 31.2. Because Q ratios of 1.0 represent equal input to magnetic anomalies from induced and remanent sources, the aeromagnetic anomalies in this region are strongly influenced by the remanent directions of both these rock types. This results in magnetic anomalies which are mapped as magnetic lows (below background) and are prevalent over the anorthosites as well over the Tellnes norite itself although to a lesser degree. The lack of a distinct strong negative anomaly over the Tellnes norite is partly due to the greater distance the source body was from the magnetometer because it was not possible to fly a draped survey in the mine area (McEnroe *et al.* 2001) due to the mountainous terrain. In addition, the Tellnes norite intrudes an area dominated by high remanence and steep negative inclinations.

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4.3.3 Demagnetization studies

Samples from the Tellnes norite, the Åna-Sira anorthosite and the Garsaknatt leuconorite were subjected to stepwise demagnetizations techniques, using both alternating field and thermal methods. A single component of magnetization is clearly apparent after a removal of a minor component allowed for easily determined directions on detailed demagnetization. Distinct behaviours were readily observed in all three sets of samples,

Tellnes samples showed three distinct behaviours during demagnetization observable in intensity loss during alternating field demagnetization (Fig. 5). The first group was representative of rocks with high coercivity, extremely hard remanence and low susceptibilities (<0.01 SI). In AF demagnetization these samples showed no or little loss of intensity up to fields of 100 mT (Fig. 6b). A mean destructive field (MDF) is not observed at these levels of demagnetization, and directions change little, if at all, during the demagnetization steps (Fig. 6b). Thermal studies on paired cores give similar directions (Fig. 6a) and show no or little loss of intensity until temperatures equal to or greater than 550 °C. Samples in this category generally have low susceptibilities, and oxide mineralogy dominated by hemo-ilmenite. This unblocking temperature would represent a Ti content of a few per cent in the magnetite, or approximately 16 per cent Ti in the haematite lamellae. These contents are well within the range of microprobe results reported by McEnroe *et al*. (1996).

A second set of demagnetization behaviour is observed in samples from the mine area that contain some magnetite. Sample susceptibilities are largely reflecting the presence of magnetite (average 0.04 SI). On AF demagnetization considerable intensity is lost at low field levels with MDFs of 20 mT or less. However, after continued AF demagnetization there is still approximately 10–20 per cent of the magnetization left (Fig. 6d). Directions may show minor variations at low demagnetization levels, but straight-line decay to the origin above the MDF. Thermal demagnetization reveals little intensity loss until after 550 °C, with small (<10 per cent) but significant amounts between 575 and 600 °C (Fig. 6c). Directions show small variations at low temperature, then consistent directions through to high temperatures. Both AF and thermal demagnetization have similar directions.

A third class of demagnetization behaviour is recognized in some Tellnes samples with low MDFs (<20 mT) and complete decay of intensity after levels of 100 mT (Fig. 6f). These sites usually have large initial intensities (>10 A m^{-1}) and show only minor overprints. Thermal demagnetization of these samples have similar behaviour to the second group, with only minor variation and no loss of intensity until temperatures above 550 °C (Fig. 6e). Again, samples demagnetized using AF and thermal produce the same characteristic directions during the cleaning process.

Anorthosite samples from the Åna-Sira in general behave similar to the first grouping of Tellnes norite samples. These show high coercivity and hard remanence on AF demagnetization indicating little or no magnetite in the sample (Fig. 7b). Thermal demagnetization reveals some remanence left above 550 °C, but directions are uni-vectoral throughout the progressive steps (Fig. 7a). Directions are consistently steep negative with west to northwest declinations. Similar directions and behaviours are seen for samples from the Garsaknatt body (Figs 7c and d). It is important to note that all samples, including the Tellnes norite, the Åna-Sira anorthosite and the Garsaknatt leuconorite have characteristic directions of steep negative inclinations and west to northwest declinations.

4.3.4 Palaeomagnetic directions and poles

A total of 32 sites, 10 from the Tellnes norite, 12 from the surrounding Åna-Sira anorthosite, and 10 from the Garsaknatt leuconorite, yield well grouped (a95 < 15)° palaeomagnetic directions (Table 1). Site means are plotted on a stereonet (Fig. 8) for each formation with associated circles of 95 per cent confidence. Tellnes norite sites have negative inclinations and declinations in the west to northwest, with a mean position of $I = -74.8°$, $D = 292.6°$, with a95 = 12.1°. Sites from the surrounding Åna-Sira anorthosite show steeper negative inclinations and declinations ranging from southwest to northeast. The mean direction for this population is found at $I = -81.2°$, $D = 326.3°$ and a95 = 6.7°. The separate Garsaknatt leuconorite yields mean directions of $I = -73.1°$, $D = 312.7°$ and a95 = 4.7°. Corresponding poles are given in Table 1. Demagnetized directions for all three units (Fig. 8) show similarity to the original NRM directions (Fig. 4) emphasizing the single component nature of these rocks.

To further constrain the palaeomagnetic data we collected three sites in the Lomland dyke, a prominent late-stage jotunite dyke cross-cutting the Egersund–Ogna anorthosite and BKS layered intrusion (Fig. 1a). This body has been dated at 916 ± 9 Ma based on U–Pb SIMS dating on zircons (Vander Auwera *et al.* 2011). Samples showed scattered directions for each site resulting in large, up to 65°, cones of confidence although mean directions are similar to the other samples studied here (Table 1). Additionally, three sites were collected from a large dolerite dyke and splays, part of the Egersund dyke swarm (Bingen *et al.* 1998), that cuts the Tellnes norite. The Egersund dykes have been investigated by Walderhaug *et al.* (2007) revealing palaeomagnetic directions significantly different from the 300 Myr older anorthosites and Tellnes norite they intrude, with positive steep inclinations and southeast declinations. Two of the Egersund dyke sites which cut the Tellnes norite produced directions similar to those obtained by Walderhaug *et al.* (2007) while the third site, produced only scattered directions (Table 1).

5 DISCUSSION

The rocks of the Tellnes norite yield strong and consistent palaeomagnetic signatures, with negative steep inclinations and northwesterly declinations. There are only minor overprints observed, and NRM directions are commonly very similar to demagnetized directions. Magnetic mineralogy is dominated by hemo-ilmenite as the main magnetic carrier, although magnetite, in varying amounts, is common in many samples. It is noted that directions become steeper with AF and thermal demagnetization implying that magnetite carries a slightly shallower direction than the hemo-ilmenite. In the surrounding rocks, the Åna-Sira anorthosite and the near-by Garsaknatt leuconorite, also have stable directions, and, likewise, the remanence is mostly carried by hemo-ilmenite magnetic mineralogy. Åna Sira directions are steeper, though generally similar to the Tellnes norite.

5.1 Reliability of directions

Van der Voo (1990) suggested a method of investigating the quality of palaeomagnetic studies presenting directions and pole positions. Studies were considered for seven criteria, with a simple yes or no response (1 or 0), with a resulting Quality (Q) value; a perfect score of 7 was possible but rare. This method was then updated and modernized by Meert *et al.* (2020) to reflect advances in palaeomagnetic and geochronologic methods. The new term, Reliability (R), is now

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used to determine adherence to seven criteria considered invaluable to the overall integrity of a given pole position.

It is advantageous to apply these criteria to the mean poles determined for the Tellnes norite, the Åna-Sira anorthosite and the Garsaknatt leuconorite. The details for the three units are discussed above, and summarized here. The first of the criteria considers a well-determined rock age and corresponding remanence age. Here, the Åna-Sira anorthosite and Tellnes norite have published radiometric ages; as these are crystallization ages, the actual remanence is gained at a somewhat younger age, as outlined below. All studies pass the second criterion as to techniques and statistical analysis with adequate site and sample numbers, precision parameters values, extensive thermal and alternating field demagnetization and tests for averaging out palaeosecular variation (PSV). This later test is performed using the technique of Deenen *et al.* (2011) to determine a confidence envelope around the pole mean; all three units produce results that indicate PSV has been averaged out in their populations.

The third criterion is an evaluation of remanence carriers, which is presented here in detail in Section 3, emphasizing the role played by hemo-ilmenite in the remanence values of these rocks. The fourth criterion concerns field tests supporting the remanence measured, and while several of these tests (fold/tilt tests and conglomerate tests) are not germane to massive intrusive bodies, the baked contact test was applied in a modified form. The ∼616 Ma Egersund dykes intrude both the Åna-Sira anorthosite and Tellnes norite, and have palaeomagnetic directions significantly different from the RIC units (Bingen *et al.* 1998; Walderhaug *et al.* 2007). One additional site in the Egersund dyke, intruding the active mining area of the Tellnes norite, has a positive inclination and northeast declination compared to the steep negative inclinations and west–northwest declinations in the norite (Table 1). The fifth criterion concerns structural and tectonic coherence and for the RIC this is evident in the lack of younger deformation in the area and the absence of more recent faulting or disruption to their geographic position.

The lack of reversals is an obvious characteristic of this and other data from the RIC, so the sixth criterion, presence of reversals, is not met. The final criterion concerns resemblance of the determined directions to younger poles for the same craton. Looking at apparent polar wader paths (APWPs) for Baltica (Bazhenov *et al.* 2016; Torsvik *et al.* 2021) indicates no intersection of the poles from the RIC to later poles, although lack of data for the late Proterozoic and early Palaeozoic from Baltica leads to pole paths that are not well determined at these times. Thus, the studies reported here are able to meet at least 5 of the 7 criteria presented by Meert *et al.* (2020) and attest to their usefulness to the interpretation of palaeomagnetic directions for Baltica at ∼900 Ma

5.2 Timing of remanence

Determination of the age of remanence depends on the rate of cooling of the individual igneous bodies and the regional cooling history, coupled with the blocking temperature of minerals that carry the remanence. McEnroe and Brown (2000) sucessfully used this correlation to evaluated the age of remanece for the proterozoic rocks exposed in the Adirondack Mounains, USA. Here, the oldest bodies in the RIC are the anorthosites (Egersund–Ogna, Helleren and Åna-Sira anorthosites) that intruded into the c.1010–970 Ma metamorphosed migmatic gneisses (Bingen et al 2021). How long the RIC anorthosites remained warm after emplacement at c. 932 Ma is not resolved. However large, km-size blocks of anorthosite are mapped in the western side of the BKS layered intrusion indicating the anorthosite was no longer a crystal mush when the BKS intruded (Marker 2013). The last Sveconorwegian magmatism in the RIC occurred with the intrusion of the smaller Tellnes norite and the jotunite dykes, such as the Lomland dyke. Zircon and baddeleyite data from these bodies indicate emplacement between 920–916 Ma, respectively (Schärer *et al.* 1996; Vander Auwera *et al.* 2011).

In all thin sections from the Tellnes norite, the Åna Sira anorthosite and the Garsaknatt leuconorite hemo-ilmenite is present. Magnetite is common in Tellnes samples, and to a much lesser extent in the Garsaknatt and Åna Sira samples. Many samples have susceptibility values < 0.005 indicating only a small amount of magnetite.

In the samples discussed here, the magnetite grains are commonly present as large discrete grains ($> 100\ \mu$m) with only minor microstructures. Given the near end-member compositions of the magnetite from Tellnes samples (McEnroe *et al.* 1996) it would have acquired a magnetization slightly below 570 °C. However, the co-existing hemo-ilmenite would not acquire a remanence until later. Based on the ilmenite-haematite 1-atmosphere phase diagram (Burton 1991; Harrison 2006; Ghiorso & Evans 2008) exsolution from a ferrian ilmenite in the composition ranges of interest exsolution would not occur until the system had cooled to below 520 °C, when the first haematite lamellae could have exsolved from ilmenite (McEnroe *et al.* 2002; 2007). With further cooling, we interpret the exsolution process would continue until it is energetically unfavourable for lamellae to form in the haematite-ilmenite system. We estimate that exsolution lamellae could have continued to form at least to 400 °C. Brown & McEnroe (2012) discussed the blocking temperatures in samples with the haematite-ilmenite exsolution and how exsolution effects the age of remanence in the Grenville Adirondack rocks. Here as in that study, our localities have different magnetic oxides that result in magnetizations acquired at different temperatures in different parts of the RIC. The hemo-ilmenite samples acquired their remanence as a thermochemical remanence (TCRM) when the lamellae form during regional cooling.

Because the RIC is a dry igneous complex there is a lack of age data from biotite, hornblende, or muscovite that could be used to construct a cooling curve in the temperature range of interest for the magnetic mineral blocking temperatures. The age data available is related to crystallization ages, which have closure temperatures much higher than the blocking temperature of magnetite, or the exsolution temperatures of the haematite lamellae in ilmenite. However, the relative order of the igneous events is well dated with c. 15 Myr separating the oldest rocks, the anorthosites, from the youngest rocks, the Tellnes norite and Lomland dyke. Present age data has the RIC emplaced in a relatively short time interval between c. 932 and 916 Ma (Bingen et al 2021, and references therein). The magnetization would have locked in post emplacement and the timing would be related to the cooling rate. There can be differences in the cooling history for the anorthosite bodies, and depending how close sites are to the BKS layered intrusion. The Garsaknatt leuconorite, a small body by surface expression, intrudes into the cooler gneisses east of the massive BKS, and thus may have cooled slightly faster than the layered intrusion.

Based on this relative cooling sequence for the RIC we estimate the minimum timing of blocking of theTCRM directions to be c. 910 Ma for the anorthosites, and for the later BKS, Tellnes and Garsaknatt intrusions to be c. 900 Ma. Though the BKS is a large intrusion and records a longer time interval for blocking magnetization it should be noted that the early part (more primitive magmas)

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of the BKS would have blocked at lower temperatures than the later evolved magmas. This is because the primitive magmas except for subunit MCU1Be, contain only hemo-ilmenite, which has a lower blocking temperature than the co-existing magnetite which occurs in the more evolved magmas (McEnroe *et al.* 2009; Brown & McEnroe 2012) and the shallower structure of the primative part of the BKS (Lee *et al* 2024). Therefore, we estimate an age of c. 900 age for the BKS. Though the Tellnes norite was emplaced after the BKS it would have acquire most of its remanence at a similar time interval because the BKS layered intrusion, due to its size, would have remained warm for some million years. We estimate that the majority of remanence, as a TCRM, in the Tellnes norite was due to the formation of lamellae in the hemo-ilmenite and this would have occurred at temperatures below 520 °C. Therefore, we do not expect a large difference in the palaeopoles from the BKS and the Tellnes norite. The 920 Myr age of the Tellnes norite is determined from zircon which has a closure temperature of 900 °C (Lee et al 1997). Thus the age of the rock was acquired at 900 °C, however, the remanence was blocked at a temperature below 520 °C, nearly 400 °C lower than the closure temperature of zircon.

5.3 Comparison to other Rogaland directions

Early palaeomagnetic studies revealed the steep reversed signatures in the rocks of the RIC (Stern & Piper 1984). More recent detailed magnetic studies on the anorthosites and layered intrusions also show these steep negative inclinations for the Egersund–Ogna anorthosite (Brown & McEnroe 2004) and the BKS layered intrusion (Brown & McEnroe 2015; Table 2). The Egersund–Ogna and Åna-Sira anorthosites show steeper inclinations than the other three bodies, with the BKS intrusion, the Garsaknatt leuconorite and the Tellnes body indicating shallower, but similar directions. *F*-test determinations (Butler 2004), comparing the combined two anorthosites to the BKS body, show the two populations are different at the 9 per cent level. This agrees with intrusion information and age determinations on the bodies with the anorthosites emplaced ∼932 Ma, the BKS layered intrusion intruding into the anorthosites at about the same time as the intrusion of the Tellnes norite with an age of ∼920 Ma. There is no age data for the Garsaknatt leuconorite, however the palaeomagnetic data, geophysical modelling (Lee et al 2024) and compositional data imply it is not directly related to the nearby Åna-Sira anorthosite, but is a later intrusion. The Tellnes norite shows the most scatter,

With a 95 per cent confidence limit of 10.6° on the magnetic directions, and when it is considered alone, does intersect with the earlier anorthosites. Taken as two distinct groups, the older anorthosites (Egersund–Ogna and Åna-Sira) and the younger BKS layered intrusion, Garsaknatt leuconorite and Tellnes norite, distinct directional differences are seen (Fig. 9 and Table 2).

Palaeolatitudes for the Tellnes norite and the Garsaknatt leuconorite are similar at −57°, with the Åna-Sira anorthosite yielding a steep latitude of −73°. These values agree well with earlier data from the BKS layered intrusion (−59°) and the Egersund–Ogna (−71°) (Table 2). The difference between the earlier anorthosites, and the later BKS layered intrusion, Garsaknatt leuconorite, and Tellnes norite intrusion is evident, with ∼12° between the earlier anorthosites, and the later layered intrusion, the Garsaknatt leuconorite, and finally the Tellnes norite. The anorthosites, at a higher palaeolatitude, indicate Rogaland was moving north towards equatorial regions with the pole plotting in a shallower position, although still relatively steep, during the cooling and remanence acquisition of all the units. Using the age difference between the Egersund–Ogna anorthosite and the Tellnes body of ∼12 Myr, a plate motion in the early Neoproterozoic of ∼10 cm yr^{-1} is suggested. Compared to current global plate velocities (Zahirovic *et al.* 2015), this implies a rather fast but acceptable plate motion for Baltica.

Although the intrusive bodies in the RIC split into two distinct groups using crystallization ages and palaeomagnetic directions, the differences are small when considered on a global scale. A plot of all the 111 sites from the five bodies (two anorthosites, Egersund–Ogna and Åna-Sira; the BKS layered intrusion; the Garsaknatt leuconorite; and the Tellnes norite), shown in Fig. 10, yields a representation of time-averaged directions from the RIC, with a mean direction of $I = -75.1°$ and $D = 307.8°$ and an $\alpha 95$ of 2.6°. This direction represents the magnetic field over a longer time period as the intrusions were emplaced at various depths and cooled to temperatures below 570 °C–520 °C over a period of at least 15 Myr. The resulting averaged pole for RIC is at 38.6° S and 212.8° E with an A-95 of 4.2°. Using the method of Deenen et al. (2011), as with the individual RIC bodies above, the determined A-95 indicates secular variation has been averaged out in this population.

5.3 Pole positions and Baltica in the late Proterozoic

The position of Baltica within the supercontinent Rodinia has been subject to several variations. The classical configuration (Li *et al.* 2008) has Baltica situated near Amazonia and Laurentia, with the southwestern part of Baltica (southern Norway and Sweden) facing Amazonia and the northern Appalachians of Laurentia, all of which are positioned at high southern latitude. Variations on positioning of Baltica and the other continents abound, but with consistent high latitude locations for Baltica (Evans 2013). Likewise, the tectonic setting of southwest Baltica has several interpretations, from the traditional collisional view of Baltica with Laurentia and Amazonia causing the Sveconorwegian orogeny (Bingen *et al.* 2021) to an accretionary boundary and a wide backarc environment (Slagstad *et al.* 2020). In all scenarios the orogenic activity, active from ∼1000 to 900 Myr, ends with the emplacement of the RIC. And, in all cases, the positions, particularly in terms of latitude, are controlled by palaeomagnetic studies of area rocks. Current poles positions published for rocks from southern Norway and Sweden from 1000 to 900 Myr are listed in Table 3, with the one caveat being that at least 8 sites must be used in calculating poles, following reliability criteria of Meert *et al.* (2020) and the need to avoid virtual geomagnetic poles influencing the pole paths and comparisons. As with many studies in orogenic belts with recently metamorphosed, intrusive and extrusive rocks, exact age determinations are often lacking, as well as definitive remanence acquisition ages. Included in Table 3 are ages reported from the original publications. Shallow poles are seen for the older units (∼1000 to ∼940 Myr) with steeper poles evident for remanence acquired at ∼900 Myr.

Apparent polar wander paths for Baltica are present by Gong *et al.* (2018, 2019), Salminen *et al.* (2021) and Kulakov *et al.* (2022); all three use very similar pole collections with ∼10 published poles to produce the paths. The paths, although different in presentation, are all very similar in form. All three reports show equatorial poles for the older units (∼1000 to ∼940 Ma), with steeper poles at the end of the Sveconorwegian orogeny (∼900 Ma), indicating considerable plate motion during this time.

The mean pole for the 111 sites from RIC provides a pole encompassing the all the units from the complex and is comparable to the 'Anorthosite Complex' pole from Stern & Piper (1984) of 44.1°

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S and 213.0° E. (Table 3). The high latitude position of southern Scandinavia agrees with other poles in southern Baltica of similar Neoproterozoic age, such as the recent pole obtained on mafic xenoliths from the neighbouring Sirdal Magmatic Belt by Kulakov *et al.* (2022). Breaking down the RIC data reported here for the older anorthosites and the younger non-anorthosite intrusions shows ∼12° of latitude difference (Table 2), and implies Baltica is moving into lower latitude position at the end of the Sveconorwegian orogeny.

The individual bodies within the RIC seem to indicate that Baltica was moving as the anorthosites and associated intrusions were emplaced and as cooling occurred to lock in the remanence. However, as with many regions in Precambrian terranes, the need for additional palaeomagnetic studies of well-dated rocks and critical determination of the age of remanence are required to fully interpret ancient plate motions.

6 CONCLUSIONS

(i) The Tellnes norite in the RIC, dated at 920 Ma, has strong remanent magnetization with steep negative inclinations and northwesterly declinations (−71.9°, 305.0°) for 10 sites collected within or near the present ilmenite mine occupying the Tellnes body.

(ii) Similar steep negative directions are found in the surrounding Åna-Sira anorthosite (−81.2°, 326.3°) into which the Tellnes norite intrudes, and to the separate, though near-by Garsaknatt leuconorite (−73.1°, 312.7°).

(iii) All three bodies have oxide mineralogy dominated by hemo-ilmenite. Magnetite is common in Tellnes norite and less abundant in the Åna Sira anorthosite and the Garsaknatt leuconorite samples. Rock magnetic and demagnetization studies indicate the remanence is dominantly held in the hemo-ilmenite.

(iv) In samples containing hemo-ilmenite remanence was not blocked in until haematite lamellae were exsolved from the ilmenite. Based on the phase diagram for the haematite-ilmenite system this would not occur until 520 °C. At higher temperatures, the ilmenite would be paramagnetic. Haematite lamellae continued to form with further cooling likely down to at least 400 °C.

(v) Comparing the three bodies studied here and two earlier studies on the BKS layered intrusion (Brown & McEnroe 2015) and the Egersund–Ogna anorthosite (Brown & McEnroe 2004) reveals statistically different directions for the older two anorthosites compared to the younger combined layered intrusion, norite and leuconorite.

(vi) Palaeolatitudes calculated for the two groups yields steeper latitudes (∼−71°) for the older anorthosites and shallower latitudes for the younger intrusions (∼−58°).

(vii) In combination with other published pole positions for Fennoscandia in the late Mesoproterozoic to early Neoproterozoic, it appears Baltica was at lower latitudes early in the time period (∼950 Ma) and then moved poleward to a steep position by 900 Ma, although it was starting to return to lower latitudes.

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DATA AVAILABILITY

Palaeomagnetic data used in this study are currently being deposited in the Magnetics Information Consortium (magic) database www2.earthref.org/magIC.

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